INCH-POUND

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MILITARY SPECIFICATION

# FLARING, FLANGING, AND BEADING MACHINES, TUBE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two types of metal tube forming machines with equipment and accessories specified herein.

1.2 Classification. The machines shall be of the following types and sizes listed in 3.5. The size to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Machine performs single flaring, beading, deburring and squaring operations on metal tubing, by means of either a rotary, or a combination of rotary and in-line motion of the forming tools against the forming dies.

Type II - Machine performs single and double flaring, beading, single and double flanging, reduction, and expansion operations on metal tubing by means of in-line ram type, motion of the forming tools, against the forming dies.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of of this document or by letter.

AMSC N/A FSC 3441

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2.1.1 Specifications standards and handbooks
cations, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS) 5801 Tabor Avenue, Philadelphia, PA 19120-5099. )

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)



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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.7 - Safety Requirements for the Construction, Care, and Use of Cold Headers and Cold Formers.

ANSI 993.114M - Pneumatic Fluid Power - Systems Standard for Industrial Equipment and Machine Tool Applications.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363. )

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, Mclean, VA 22101-4269.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(lion-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.



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3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, and repair. Type I machines shall operate by manually inserting a squared and deburred workpiece into a forming die until contact is made against an adjustable workpiece stop. The die is closed, securely gripping the workpiece. A rotating forming tool such as a flaring center, or beading tip, mounted on the spindle of the machine, is manually advanced until contact is made with the workpiece and the end form completed. After completion of the forming operation, the rotating tool is returned to its starting position, the die is opened, and the workpiece removed. Type II machines shall operate by manually positioning the squared and deburred workpiece in the open die against an adjustable workpiece stop. Actuation of the control shall initiate the automatic sequence and all operations shall occur in sequence and be performed without further operator action. Operation of the control shall close the die and grip the workpiece, move the workpiece stop out of position, and move the forming punch into concentric alignment with the tube end. Then the power head shall axially force the forming punch into the tube end forming the tube end to the configuration of the punch and die. After completion of the forming operations, the die shall open, allowing the finished workpiece to be removed.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.4.1), metric (see 6.4.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under-this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).



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3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform with all requirements of 29 CFR 1910 and ANSI B11.7. If a conflict arises between 29 CFR 1910 and B11.7, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified in detail (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machines, supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. Additional guarding dependent on workpiece size and configuration shall be provided when specified and fully described by the procuring activity (see 6.2).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine, and its components, materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. An automatic lubricating system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.



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3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated l torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components, and specific requirements.

3.4.1 Type I machine. Type I machines shall consist of not less than a base, spindle assembly, jaw assembly, lubrication system, electrical system, system, and the other components required to meet requirements specified herein. When specified (see 6.2), Type I machines shall be provided with an automatic sequence of operation control.

3.4.1.1 Base. The machine shall have a base of adequate strength to withstand forces applied while the machine is operating at its maximum rated capacity. The base shall be a workbench mounted, base cabinet style, or floor stand style, as specified and fully described (see 6.2). When machine components are enclosed within the base, base cabinet or floor stand, removable or hinged panels shall be provided to allow access to these components for lubrication, adjustment, or maintenance.

3.4.1.2 Spindle assembly. The spindle assembly shall consist of spindle housing, spindle sleeve, spindle bearings, spindle, spindle drive, and spindle advance mechanism. The spindle shall be mounted on ball or roller bearings, contained in the spindle sleeve and the spindle sleeve In the spindle housing. The spindle shall be driven by an electric motor through a belt and pulley drive. The spindle sleeve shall advance and retract the spindle through a gear segment and lever arrangement.



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3.4.1.2.1 Spindle. The spindle shall be of sufficient size and strength to withstand forces encountered while operating at the machine's maximum rated capacity. The spindle end shall be constructed to accommodate the various beading, squaring and deburring heads and flaring centers.

3.4.1.3 Jaw assembly. The jaw assembly shall accommodate dies necessary for performing singl flaring, beading, deburring, and squaring operations on tubing sizes specified in table I. The jaw assembly shall consist of one stationary jaw, and one movable jaw with a hinging and locking mechanism. The jaws shall hold the forming dies, which shall firmly hold the workpiece. The stationary jaw shall be mounted to the base of the machine. The movable jaw shall hinge to the stationary jaw, and shall open sufficiently to allow for inserting and removal of the workpiece.

3.4.2 Type II machine. Type II machines shall consist of not less than a base, die holder, die closing mechanism, tool holder, tube stop, power head, pneumatic or hydraulic system, lubrication system, electrical systems and other components required to meet requirements specified herein. Type II machines shall have an automatic mode of operation. Unless otherwise specified (see 6.2), the machine shall be provided with a manual override control which overrides the automatic cycle.

3.4.2.1 Base. The machine shall have a base of adequate strength to withstand forces applied while the machine is operating at its maximum rated capacity. The base shall be a workbench mounted, base cabinet style, or floor stand style, as specified (see 6.2). When machine components are enclosed within the base, base cabinet, or floor stand, removable or hinged panels shall be provided to allow access to these components for lubrication, adjustment, or maintenance.

3.4.2.2 Die holder. The machine shall have a die holder capable of accommodating all dies necessary for all tube forming operations shown in table II. The die holder shall facilitate changing of the dies and shall securely hold the die in axial alignment with the punches.

3.4.2.3 Die closing mechanism. The machine shall have a die closing mechanism. that will open the die set for removal or insertion of the workpiece, and close the" die set, securely gripping the workpiece. The die closing mechanism shall provide adequate clamping force, to prevent the workpiece from slipping during the forming operation. The mechanism shall be powered by either a hydraulic system comforting to 3.4.3, or a Pneumatic system conforming to 3.4.4.

3.4.2.4 Tool holder. Machines supplied with single workstations shall have a multiposition tool holder which supports not less than two forming tools. The tool holder may be either the sliding or rotating type, and shall successively align the forming tools with the forming die.

3.4.2.5 Tube stop. The machine shall have an adjustable tube stop for positioning the end of the tube in the proper position to receive the forming tools. The tube stop may be either mounted in the tool holder or it may be the swing away type.

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TABLE I

SIZES AND CAPACITIES

| TYPE I | | | | | |
|---|---|---|---|---|---|
| METAL TUBE FORMING OPERATION | Size 1 | Size 2 | Size 3 | Size 4 | Size 5 |
| Single Flaring, Squaring and Deburring, diameter, inches | 1/8 to 2 | 1/8 to 3 | 1/2 to 4 | 1-1/4 to 6 | 1-1/4 to 8 |
| Beading diameter, inches | 1/4 to 1-1/2 | 1/4 to 3 | 5/8 to 4 | 1-1/4 to 6 | 1-1/4 to 8 |
| Motor H. P. (H. P. of spindle motor or hydraulic pump motor). | 1/3 | 3/4 | 2 | 2 | 2 |

TABLE II

SIZES AND CAPACITIES

| TYPE II | | | | | | |
|---|---|---|---|---|---|---|
| METAL TUBE FORMING OPERATION | Size 1 | Size 2 | Size 3 | Size 4 | Size 5 | Size 6 |
| Single & Double Flaring, Single & Double Flanging, Expansions & Reductions, diameter, inches | 1/8 to 3/4 | 1/8 to 1 | 1/4 to 1-1/2 | 1/4 to 1-1/2 | 1/4 to 2-1/2 | 1/4 to 3-1/2 |
| Beading, diameter, inches | 1/8 to 3/4 | 1/8 to 1 | 1/4 to 1-1/2 | 1/4 to 1-3/4 | 1/4 to 2-1/2 | 1/4 to 3-1/2 |
| Motor H. P. (H. P. of hydraulic pump motor, not applicable for pneumatic machine, on shop air) | 5 | 7-1/2 | 10 | 10 | 15 | 15 |

NOTE: Type I and Type II, requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure (or less) to the larger figure (or more).





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3.4.2.6 Power head. The machine shall have a power head which applies axial force to the forming tools, and forces the tool into the workpiece to form the end connection. The power head shall be operated by either a pneumatic system conforming to 3.4.3, or a hydraulic 'system conforming to 3.4.4. The power head shall exert force through a straight inline ram, an impact piston, or through a toggle arrangement, and shall be capable of applying adequate force to perform forming operations on tubing of the sizes specified in table II. Means shall be provided for adjusting the amount of force exerted by the power head.

3.4.3 Pneumatic system. The pneumatic system when furnished, shall conform to the requirements of ANSI B93.114M, and operate on a maximum air pressure requirement of 90 psig. The system shall have an air line filter, water trap, and oil mist lubricator, as necessary. The system shall include all devices such as pressure regulators, pressure reducers, solenoid valves, and check valves that are necessary to meet the requirements specified herein. Air pressure gauges shall be provided to indicate input air pressure in all lines that are downstream from a pressure control device.

3.4.4 Hydraulic system. The hydraulic system when furnished shall conform to or exceed the requirements of JIC H-1 and shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, provisions for draining, and a cleanout access plate.

3.4.5 Electrical system. The electrical system shall be wired to conform to NFPA 73. Unless otherwise specified (see 6.2), each machine shall be wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-Hertz (Hz) circuit. Unless otherwise specified (see 6.2), the machine shall be initially wired for operation on 460 volts.

3.4.5.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), each motor shall have a dripproof enclosure.

3.4.6 Tooling. Tooling shall be furnished as specified and fully described (see 6.2). Tooling such as punches, dies, beading tips, rollers, and flaring centers shall be made to resist wear and deformation normally encountered during machine operations. The tooling shall produce the accuracies necessary to provide tube end connections to the dimensional accuracies specified in 3.6.1 and 3.6.2.

3.4.7 Morklight. When specified (see 6.2), a worklight shall be provided. The lights have a protective shield, an on-off switch and shall be adjustable to reflect maximum light on the work area. If a built in lighting system is provided, the adjustable light described above shall not be required.

3.4.8 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the



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nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall be not less than specified in tables I & II for the machine size specified.

3.6 Performance. The machine shall be capable of meeting the requirements of 3.6.1 through 3.6.4.

3.6.1 Type I machine. Type I machines shall be capable of producing single flares and beads on ferrous and non-ferrous tubes of the sizes specified in table I. Type I machines shall also be capable of squaring and deburring all tubes of the above material and sizes. The accuracy of the end connections shall conform to 3.6.3. The speed of operation shall conform to 3.6.4.

3.6.2 Type II machine. Type II machines shall be capable of producing single flares, double flares, beads, and 90-degree single and double-lap flanges on ferrous and non-ferrous tubes of the sizes specified in table II. The accuracy of the end connections shall conform to 3.6.3. The speed of operation shall conform to 3.6.4. Type II machines shall also be capable of forming expansions and reductions on tube ends.

3.6.3 Accuracy. The accuracy of the end fittings produced, including single flares, double flares, beads, and flanges shall be square to the wall of the tube within one degree, and concentric with the outside diameter of the tube within 0.005 inch.

3.6.4 Speed of operation. When using squared and deburred tubing of Type 304 or 316 cold drawn, annealed, seamless, stainless steel tubing, the machine shall be capable of producing five samples of the same size, (of any size within the range specified in tables I & II), within the time period specified below:

| | | |
|---|---|---|
| Single flares | - Five single flares in not more than 60 seconds | |
| Beads | - Five beads in not more than 80 seconds | |
| Double flares | - Type II machine only | |
| Expansion | - Type II machine only | Five forming operations in not |
| Reductions | - Type II machine only | more than 80 seconds |
| Flanges | - Type II machine only | |

NOTE: The above time limitations shall exclude squaring & deburring time; however, limitations shall include workpiece handling and actual forming time of the workpiece.

3.6.5 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally furnished with the manufacturer's commmercial machine shall be furnished.



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3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.

3.10 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in bold face on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate or chart. When specified (see 6.2), a lubrication plate or chart shall be attached to each machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2), the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)
Viscosity

3.10.2 Name late. A nameplate shall be securely attached to each machine. Unless oterwise specified (see 6.2), the nameplate shall contain the information listed below:

Nomenclature
Manufacturer’s name
Manufacturer’s model designation
Manufacturer’s serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.10.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Type of fluid (SAE number or military or Federal specification number)
Type of filter

3.11 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the first article examination and all tests, to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination-in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. The machine shall be examined to determine compliance with all requirements of this specification.



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4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 410 to 1040 Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.7.2 Operational test. The machine shall be operated at no load for not less than 15 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Performance tests. Type I machines shall be subjected to the test specified in 4.7.3.1 Type II machines shall be subjected to the test specified in 4 7.3.2. Type I and Type II machines shall, in addition, be subjected to the beading accuracy and speed test specified in 4.7.3.3.

4.7.3.1 Single flaring accuracy and speed (Type I machines only). Type I machines shall be tested for single flaring accuracy and speed by producing fifteen single flare samples (in squared and deburred tubing) meeting the requirements of AISI Type 304, or Type 316 cold drawn, annealed, seamless, stainless steel tubing. The fifteen samples shall consist of:

a. Five samples in tubing of the smallest size specified in Table I.

b. Five samples in tubing of the largest size specified in Table I.

Five samples in tubing of a size equal to the average of the size range specified in Table I.

During the above test, the time required to produce each 5 sample groups shall be determined. All of the flares produced shall be measured for accuracy. All of the flares shall conform to the accuracy requirements of 3.6.3. The time required to produce each group of five flares, shall conform to the requirements of 3.6.4.

4.7.3.2 Tube forming accuracy and speed (Type II machines only.) Type II machines shall be tested for tube forming accuracy and speed, in accordance with 4.7.3.1, and shall meet the requirements of table II, (except that double flares, expansions, reductions, and flanges shall be produced in lieu of the single flares specified in 4.7.3.1). All of the formed tubes shall conform to the accuracy requirements of 3.6.3. The time required to produce each group of 5 formed tubes shall conform to the requirements of 3.6.4.

4.7.3.3 Beading accuracy and speed. Type I and II machines shall be tested for beading accuracy and speed, in accordance with 4.7.3.1, (except that the beads shall be produced in lieu of the single flares specified in 4.7.3.1). All of the beads shall conform to the accuracy requirements of 3.6.3. The time required to produce each group of five beads shall conform to the requirements of 3.6.4.

4.7.3.4 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.



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4.8 Packaging Inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Machines covered by this specification are intended for use in forming tube end connections conforming to Military Standards requirements and tolerances.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify system of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5.).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. When automatic operation sequence is required, specify and fully describe (see 3.4.1).

k. If machine floor stand is required, specify and fully describe (see 3.4.1.1).

1. Machine controls, if different (see 3.4.2).



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m. If machine floor stand is required, specify and fully describe (see 3.4.2.1).

n. Electrical system, if different (see 3.4.5).

o. Initial input voltage, if different (see 3.4.5).

p. Motor, if different (see 3.4.5.1).

q. When tooling is required; specify type, size, and quantity and fully describe (see 3.4.6).

r. Worklight, if required (see 3.4.7).

s. Hour meter, if required (see 3.4.8).

t. Size and capacity, if different (see 3.5).

u. Additional performance features, if required (see 3.6.5).

v. Specify and fully describe optional equipment, if required (see 3.8).

w. Specify applicable repair parts, if required (see 3.9).

x. Lubrication plate, if required (see 3.10.1).

y. Specify information to be provided on lubrication plate (see 3.10.1).

z. Nameplate, if different (see 3.10.2).

aa. Hydraulic fluid requirements plate, if different (see 3.10.3).

bb. Inspection conditions, if different (see 4.3).

cc. First article inspection, if different (see 4.4).

dd l Quality conformance inspection, if different (see 4.5).

6.2.1 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements , a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.



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| Reference Paragraph | DID Number | DID Tiltle | Suggested Tailoring |
|---|---|---|---|
| . | . | . | . |
| . | . | . | . |
| . | . | . | . |

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List. (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.2.2 Technical data. All technical data furnished should be written in the English language.

6.3 First article. When first article inspection is required, the contracting oft leer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that-the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.4.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.5 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.


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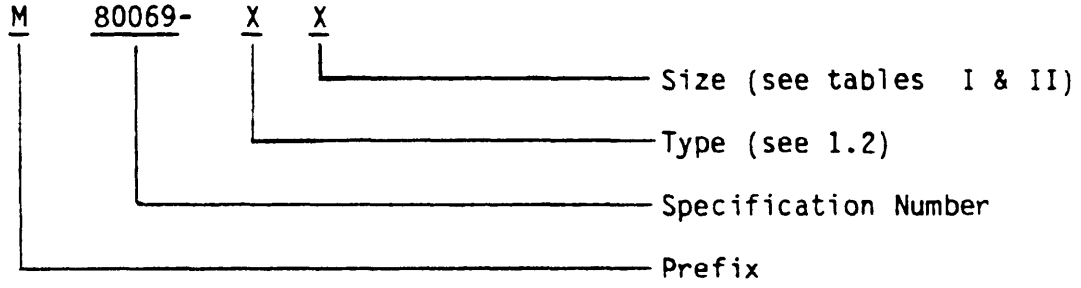
6.7 Safety and health requirements. The 29 CFR 1910 limits only the total hazard 1evel (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment-in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.8 Training. Training required by the procuring activity should be provided as specified in the contract.

6.9 Warrant Warranty requirements should be as specified by the procuring activity in the contract.

6.10 Inspection location. The contractor shall identify in his response to the solicitation the location where inspection and tests are to be performed.

6.11 Part or Identifying Number (PIN). The PIN to be used for machines acquired by this specification are created as follows:




6.12 Subject term (keyword) listing.

Beading
End Preparation
Flanging
Flaring
Tubing

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

Project (3441-0126)

Review activities:
Air Force - 84

User activities:
Navy - MC, YD


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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-F-80069D | Flaring, Flanging, and Beading Machines, Tube |

| a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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